

User Research Report

Role 1: Tourist

Description Users who interact with the system as Tourist

Persona 1

Name: Priya Sharma

Age: 28

Occupation: Graphic Designer

Location: Mumbai, India

Goals: To confidently and safely explore remote and scenic locations in Northeast India as a solo traveler., To have a reliable and quick way to alert authorities in case of an emergency., To provide peace of mind to her family by allowing them to see her location when she is in remote areas., To stay informed about potentially unsafe or restricted zones in real-time.

- To confidently and safely explore remote and scenic locations in Northeast India as a solo traveler.
- To have a reliable and quick way to alert authorities in case of an emergency.
- To provide peace of mind to her family by allowing them to see her location when she is in remote areas.
- To stay informed about potentially unsafe or restricted zones in real-time.

Key Characteristics

- Tech-savvy and comfortable using smartphone apps for travel and safety.
- Adventurous and enjoys off-the-beaten-path travel experiences.
- Safety-conscious but values her privacy and autonomy.
- Wants a seamless travel experience without the hassle of carrying multiple physical documents.

Pain Points

- Anxiety about personal safety when traveling alone in unfamiliar regions.
- Worrying her parents who are concerned about her well-being during solo trips.
- Fear of not being able to communicate her exact location quickly in an emergency, especially in areas with poor network.
- Uncertainty about local conditions and which areas might be restricted or high-risk.
- The risk of losing important physical documents like her ID card while traveling.

Empathy Map

Thinking: I really hope this app's panic button is easy to access and works instantly, even with a poor signal. I'm

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okay with sharing my location for safety, but I value my privacy—I don't want to feel like I'm being constantly monitored. Will this app drain my phone battery? I need it to last all day.

Seeing: I see stunning photos of remote mountains and valleys on social media that I want to explore. I also see news reports about tourist safety incidents, which makes me cautious. I see my smartphone as my lifeline for navigation, communication, and safety.

Saying: "I'll share my live location with you, Mom, so you don't have to worry while I'm on my trek." "Is there an app that can alert me if I accidentally wander into a restricted area?" "Having a secure digital ID on my phone would be so much easier than carrying my passport everywhere."

Doing: I'm researching safety tips for solo female travelers in Northeast India. I'm downloading and setting up the VeriTour app before my trip. I'm carefully planning my itinerary and sharing it with my family. I'm enabling opt-in location sharing for a specific hike.

Hearing: I hear my parents' voices telling me to be extremely careful and to call them every day. I hear friends telling me how brave I am to travel alone. I read reviews from other travelers about the best and safest places to visit.

Feeling: I feel excited and adventurous about my upcoming solo trip, but also a little anxious about my safety. I feel empowered by my independence, but I also want the reassurance that help is just a button-press away. I feel confident when I have the right technology to keep me safe and my family informed.

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Role 2: Police Officer

Description Users who interact with the system as Police Officer

Persona 1

Name: Inspector Arjun Singh

Age: 42

Occupation: Police Inspector, Tourist Police Unit

Location: Shillong, Meghalaya

Goals: To rapidly assess emergency alerts and dispatch the nearest response team effectively., To reduce the time spent on manual paperwork, especially for filing missing person FIRs., To have a single, reliable source of truth for a tourist's identity, location, and itinerary during an incident., To proactively identify and monitor potential high-risk zones for tourists to prevent incidents before they happen.

- To rapidly assess emergency alerts and dispatch the nearest response team effectively.
- To reduce the time spent on manual paperwork, especially for filing missing person FIRs.
- To have a single, reliable source of truth for a tourist's identity, location, and itinerary during an incident.
- To proactively identify and monitor potential high-risk zones for tourists to prevent incidents before they happen.

Key Characteristics

- Tech-savvy enough to use modern software but requires interfaces to be extremely intuitive and fast, especially under pressure.
- Dedicated to public safety and feels a strong sense of responsibility for the well-being of tourists in his jurisdiction.
- Accustomed to a high-stress, time-critical work environment where quick decisions are paramount.
- Methodical and detail-oriented when it comes to incident logging and investigation.

Pain Points

- Receiving delayed or incomplete information about a tourist in distress, leading to slower response times.
- The cumbersome and time-consuming manual process of filing a First Information Report (FIR).
- Lack of real-time location data for tourists, especially in remote areas or places with poor network coverage.
- Difficulty coordinating with the tourism department during emergencies due to siloed communication channels.

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- Wasting critical resources responding to false alarms or unverified information.

Empathy Map

Thinking: This is so much better than the old way. I have all the critical information in one place. Every second we save could be the difference between life and death. I hope the network in the remote areas is strong enough to keep the data flowing. We have a huge responsibility to use this data ethically and protect tourist privacy.

Seeing: I see a large screen with a real-time map of my jurisdiction. Alerts pop up with a loud sound, showing the exact location of a tourist in distress. I can see their photo, their itinerary, and their last known movements on a single dashboard, instead of shuffling through papers.

Saying: 'Dispatch the nearest unit to these coordinates immediately. I'm sending the tourist's full profile to your device now.' 'Acknowledge alert. We are responding.' 'This pre-filled E-FIR will save us hours of paperwork.' 'Is this alert verified or another false alarm?'

Doing: I am acknowledging a new panic alert on the dashboard within seconds. I am clicking on the alert to pull up the tourist's complete profile and live location. I am coordinating with my field units over the radio, providing them with real-time updates from the system. I am generating an E-FIR with a single click for a confirmed missing person case.

Hearing: I hear the distinct, urgent sound of a new alert from the system. Over the radio, I hear my team confirming, 'We have visual on the tourist, they are safe.' I hear my superior asking for a status update, which I can give instantly by looking at the dashboard. I hear a tourism official on the phone coordinating about a newly marked high-risk zone.

Feeling: I feel a surge of adrenaline and responsibility when a critical alert comes in. I feel confident and empowered because the system gives me the clear, real-time information I need to act decisively. I feel immense relief when my team reports that a tourist has been found safe. I feel frustrated when a false alarm wastes our precious time and resources.

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Role 3: Tourism Official

Description Users who interact with the system as Tourism Official

Persona 1

Name: Rajiv Mehra

Age: 48

Occupation: Director of Safety & Operations, State Tourism Department

Location: Shillong, Meghalaya

Goals: To proactively manage tourist safety by using data-driven insights to prevent incidents before they occur., To improve the state's reputation as a premier, safe tourist destination, thereby boosting tourism revenue., To efficiently allocate safety resources (like tourist police) to high-risk or high-density areas identified through real-time data., To establish a seamless, rapid communication and coordination protocol with police departments during emergencies., To analyze long-term safety data and trends to inform tourism policies and infrastructure development.

- To proactively manage tourist safety by using data-driven insights to prevent incidents before they occur.
- To improve the state's reputation as a premier, safe tourist destination, thereby boosting tourism revenue.
- To efficiently allocate safety resources (like tourist police) to high-risk or high-density areas identified through real-time data.
- To establish a seamless, rapid communication and coordination protocol with police departments during emergencies.
- To analyze long-term safety data and trends to inform tourism policies and infrastructure development.

Key Characteristics

- Data-driven and analytical; prefers evidence over anecdotes.
- Strategic thinker focused on prevention and long-term planning.
- Tech-savvy for a government official; understands the value of a centralized digital dashboard.
- Collaborative and adept at navigating inter-departmental politics, especially with law enforcement.
- Possesses a strong sense of public service and accountability for tourist welfare.

Pain Points

- Lack of a single, real-time view of tourist locations makes it impossible to get an accurate picture during a crisis.

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- Communication with police during incidents is often slow and inefficient, relying on phone calls and manual information sharing.
- Resource allocation is based on historical data and guesswork, not on live situations.
- Inability to quickly warn tourists about sudden, localized risks like landslides or civil unrest.
- A single negative safety incident receives significant media attention, damaging years of work to build a positive reputation.

Empathy Map

Thinking: I feel the weight of responsibility for every tourist's safety. I'm constantly worried that a major incident will happen on my watch and we won't be able to respond effectively. I'm frustrated with the bureaucratic delays that slow down the adoption of crucial technology. I wonder if I can get all the stakeholders, especially the police, to truly buy into a shared system. This VeriTour dashboard could finally give me the tools I need to do my job properly.

Seeing: I see daily reports on tourist numbers, but they're always outdated. I see news headlines that sensationalize any single safety incident, threatening our state's reputation. I look at static maps and try to guess where our resources are most needed. I see the potential for technology to give me a real-time, bird's-eye view of everything, but my current tools are archaic.

Saying: 'We need to move from being reactive to proactive. A unified dashboard is not a luxury, it's a necessity.' I often say, 'How can we get law enforcement the information they need in seconds, not minutes?' In meetings, I'm constantly pushing for 'data-driven resource allocation' instead of relying on last year's statistics.

Doing: I spend my days in meetings trying to foster better collaboration between my department and the police. I'm constantly analyzing past incident reports to find patterns that could have been prevented. I'm championing the VeriTour project, creating presentations and proposals to get the necessary approvals and budget. During a crisis, I'm on the phone constantly, trying to piece together information from multiple, disconnected sources.

Hearing: I hear from my team about the challenges of managing safety on the ground with limited information. I hear from police officials about the slow and inefficient communication channels during emergencies. From my superiors, I hear the pressure to boost tourism revenue, which I know is directly tied to our reputation for safety. After an incident, I hear the criticism from the media and the public.

Feeling: I feel a deep sense of accountability for the well-being of visitors to my state. I'm often frustrated by the

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limitations of our current systems, especially when I know better solutions exist. I'm hopeful and optimistic about the potential of a system like VeriTour to revolutionize how we manage tourist safety. During an emergency, I feel an immense amount of stress, knowing that every second counts.

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Role 4: Entry Point Staff

Description Users who interact with the system as Entry Point Staff

Persona 1

Name: Rajesh Kumar

Age: 42

Occupation: Immigration Officer at an International Airport

Location: Guwahati, Assam

Goals: To process and verify tourist documents efficiently and accurately to minimize queue times., To quickly and securely issue VeriTour Digital IDs to every eligible tourist., To contribute to the overall safety and positive experience of tourists visiting the region., To easily navigate the VeriTour portal without requiring extensive technical training.

- To process and verify tourist documents efficiently and accurately to minimize queue times.
- To quickly and securely issue VeriTour Digital IDs to every eligible tourist.
- To contribute to the overall safety and positive experience of tourists visiting the region.
- To easily navigate the VeriTour portal without requiring extensive technical training.

Key Characteristics

- Detail-oriented and methodical, ensuring all verification steps are followed correctly.
- Calm and efficient under pressure, especially during peak travel hours.
- Comfortable using web-based portals and computer systems for daily tasks.
- Values security protocols and understands the importance of accurate data.
- Communicates clearly and professionally with a diverse, international group of travelers.

Pain Points

- Frustration with current slow and often manual verification processes that lead to long queues.
- Stress associated with accurately handling sensitive documents and data for a high volume of people.
- Concern that a new system might be complex and slow down his workflow instead of improving it.
- Dealing with language barriers when explaining new procedures to tourists.
- Lack of integrated systems, requiring him to switch between multiple applications to complete one verification.

Empathy Map

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Thinking: I hope this new VeriTour system is truly faster. The last thing I need is another complicated process that creates longer queues. I must be extremely careful with this data; a mistake could cause a serious problem for the tourist and for security. I wish all these verification steps were on a single, integrated platform.

Seeing: I see a long line of tourists in front of my counter, many looking tired from their journey. I see my computer screen with the new VeriTour portal open next to our old immigration system. I see stacks of passports and visa documents that I need to process.

Saying: Welcome, ma'am. May I see your passport? I will now issue your secure Digital Tourist ID for your safety here. Please scan the QR code to activate it on your phone. Have a safe and pleasant trip.

Doing: I am carefully scanning each tourist's passport and visa. I am entering their details into the VeriTour portal to generate their Digital ID. I am trying to work quickly and methodically to keep the queue moving, switching between multiple screens to complete the process.

Hearing: I hear the constant hum of the airport, the announcements over the PA system, and the chatter of people in different languages. I hear my colleague at the next counter explaining the same new ID process. I hear the sound of my keyboard clicking and the stamp hitting the passports.

Feeling: I feel a sense of pressure to work quickly but also a great responsibility to be accurate. I feel frustrated when the current systems are slow and I have to make people wait. I am hopeful that this new VeriTour system will make my job easier and help keep our visitors safe, but I'm also a bit anxious about how complex it might be.

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Role 5: Government Representative

Description Users who interact with the system as Government Representative

Persona 1

Name: Rajan Verma

Age: 52

Occupation: Joint Secretary, Ministry of Tourism

Location: New Delhi, India

Goals: Successfully implement VeriTour to enhance India's reputation as a safe and modern tourist destination., Establish a clear governance and legal framework for the project, ensuring compliance with data privacy laws like DPDPA., Improve inter-departmental coordination between Tourism, Home Affairs, and External Affairs to create a unified safety ecosystem., Use data and analytics from the system to make informed policy decisions and demonstrate a measurable reduction in tourist safety incidents.

- Successfully implement VeriTour to enhance India's reputation as a safe and modern tourist destination.
- Establish a clear governance and legal framework for the project, ensuring compliance with data privacy laws like DPDPA.
- Improve inter-departmental coordination between Tourism, Home Affairs, and External Affairs to create a unified safety ecosystem.
- Use data and analytics from the system to make informed policy decisions and demonstrate a measurable reduction in tourist safety incidents.

Key Characteristics

- Policy-oriented and strategically focused on the project's long-term impact on tourism.
- Concerned with legal compliance, data governance, and national security implications.
- Data-driven, requiring clear metrics and success criteria to justify project investment and continuation.
- Adept at navigating government bureaucracy and fostering collaboration between different ministries.
- Reputation-conscious, both for the ministry and the country.

Pain Points

- Negative international press from isolated tourist safety incidents damages the national brand and tourism revenue.
- Siloed communication and data sharing between police and tourism departments lead to inefficient and slow

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incident response.

- Lack of real-time, aggregated data makes it difficult to proactively manage tourist safety risks and allocate resources.
- The legal and privacy challenges of implementing a large-scale monitoring system are complex and carry significant political risk.
- Bureaucratic hurdles in getting various state and central government agencies to adopt and integrate with a unified platform.

Empathy Map

Thinking: "A single data breach could be a political and diplomatic disaster. Are the security measures truly robust? How can I navigate the bureaucratic hurdles to get all state and central agencies to adopt this platform uniformly? This project has the potential to be a game-changer, but the implementation risks are significant."

Seeing: I'm looking at the detailed Business Requirements Document for VeriTour, focusing on the sections for legal compliance, stakeholder responsibilities, and the risk assessment, particularly the parts about data privacy and inter-agency dependencies.

Saying: "We must ensure this system is fully compliant with the Digital Personal Data Protection Act. I need to see clear metrics on how this will reduce incident response times and improve our national image as a safe tourist destination. How will we ensure seamless coordination between the police and tourism departments across different states?"

Doing: I am organizing inter-ministerial meetings with representatives from Home Affairs and External Affairs, drafting policy briefs for the project's governance framework, and consulting with legal advisors to vet the data handling protocols outlined in the BRD.

Hearing: I'm hearing concerns from my counterparts in the Ministry of Home Affairs about data access protocols, pressure from the Prime Minister's Office to showcase technological advancements in governance, and reports from international media about past tourist safety incidents.

Feeling: I feel a strong sense of responsibility for the project's success, as it directly impacts our national reputation. I'm cautiously optimistic about its potential but also pressured by the complexity of the legal compliance and the need for flawless inter-departmental collaboration.

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Role 6: Family Member

Description Users who interact with the system as Family Member

Persona 1

Name: Anjali Sharma

Age: 52

Occupation: Bank Manager

Location: Mumbai, India

Goals: Have peace of mind by being able to occasionally check on her son's location during his solo trekking trip in Northeast India., Receive immediate and actionable alerts if her son activates the panic button or is flagged by the system., Access a simple, secure, and easy-to-use web portal to view her son's shared location without needing technical expertise.

- Have peace of mind by being able to occasionally check on her son's location during his solo trekking trip in Northeast India.
- Receive immediate and actionable alerts if her son activates the panic button or is flagged by the system.
- Access a simple, secure, and easy-to-use web portal to view her son's shared location without needing technical expertise.

Key Characteristics

- Caring and protective parent, but respectful of her adult son's independence.
- Moderately tech-savvy; uses a smartphone and laptop daily for banking and communication but prefers straightforward interfaces.
- Prone to anxiety when her son travels to remote areas, especially those with known connectivity issues.
- Values security and official channels for information during a potential crisis.

Pain Points

- Worry and stress when her son doesn't respond to calls or messages for several hours.
- Feeling helpless and not knowing who the right local authorities are to contact in an emergency.
- Doesn't want to constantly bother her son by calling, which might ruin his travel experience.
- Fear of receiving delayed information in a critical situation.

Empathy Map

Thinking: I feel a bit guilty for tracking him like he's a child, but the peace of mind is worth it. What if he loses

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network signal for a long time? How will the system know if he's okay? I trust him, but I don't trust the unpredictable terrain. I'm so glad this system exists; at least I know who to contact if something goes wrong.

Seeing: I see the secure web portal on my laptop with a map showing my son's last updated location. I see news reports about the dangers of solo trekking which makes me nervous. I see notifications pop up on my phone, and I hope it's a message from him, not an emergency alert.

Saying: "I hope he remembers to keep his phone charged. I just want to know that he's safe. This tracking portal is a relief, but I hope we never need the panic button. Can you just send me one message a day so I don't worry?"

Doing: I am opening the web link every few hours to check his location on the map. I am keeping my phone close and the ringer volume on high, just in case an alert comes through. I am trying to distract myself with my work, but I keep thinking about his trip.

Hearing: I hear the constant notification sounds from my phone, and my heart skips a beat each time. I hear my husband telling me to relax and let our son enjoy his adventure. I hear my son's voice in my head, reassuring me he'll be careful before he left.

Feeling: I feel a constant mix of anxiety and reassurance. I'm worried about all the things that could go wrong in a remote area. But when I see his location update on the portal, I feel a wave of relief. I feel helpless when I can't reach him, but empowered knowing there's a formal system in place to respond in an emergency.

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Role 7: System Administrator

Description Users who interact with the system as System Administrator

Persona 1

Name: Aarav Sharma

Age: 38

Occupation: Lead Cloud & Security Administrator

Location: Bangalore, India

Goals: To maintain 99.9% uptime for the VeriTour system to ensure it's always available during emergencies., To guarantee zero security breaches and ensure full compliance with the Digital Personal Data Protection Act (DPDPA)., To implement robust monitoring and alerting to proactively identify and resolve infrastructure issues before they impact users., To automate deployment and scaling processes to handle fluctuating tourist loads efficiently., To streamline the management of secure, role-based access for Police, Tourism, and other government stakeholders.

- To maintain 99.9% uptime for the VeriTour system to ensure it's always available during emergencies.
- To guarantee zero security breaches and ensure full compliance with the Digital Personal Data Protection Act (DPDPA).
- To implement robust monitoring and alerting to proactively identify and resolve infrastructure issues before they impact users.
- To automate deployment and scaling processes to handle fluctuating tourist loads efficiently.
- To streamline the management of secure, role-based access for Police, Tourism, and other government stakeholders.

Key Characteristics

- Highly technical with deep expertise in cloud infrastructure (AWS/Azure), Kubernetes, and CI/CD pipelines.
- Extremely security-conscious, with a 'zero-trust' mindset.
- Methodical and process-oriented, especially when managing system changes and updates.
- Remains calm and analytical under pressure, particularly during critical incidents or system outages.
- Proactive problem-solver who prefers preventing issues over fixing them.

Pain Points

- The immense pressure of managing a system where downtime could directly impact someone's safety.

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- Frustration with the unreliability of external government APIs (for KYC, E-FIR) which can cause system instability.
- The complexity of managing a distributed system with microservices, AI models, and a blockchain network.
- Worrying about sophisticated cyber threats targeting sensitive tourist PII and location data.
- Being on-call 24/7 and the challenge of diagnosing issues across a complex technology stack late at night.

Empathy Map

Thinking: This system's uptime is directly linked to someone's safety; there is zero room for error. I hope our fallback mechanism for the government APIs is robust enough for the upcoming tourist season. Are we truly DPDPA compliant with how we're handling off-chain PII storage? I need to think about how to better protect us from a sophisticated DDoS attack.

Seeing: A wall of monitors showing Grafana dashboards with real-time system metrics, Kubernetes pod statuses, and security alerts from our SIEM. I'm constantly watching latency graphs, error rate counters, and CI/CD pipeline logs. I also see a backlog of Jira tickets for access requests and infrastructure changes.

Saying: "I'm seeing a latency spike on the KYC verification API again. We need to check the status of that external government service. Let's double-check our firewall rules and access policies before onboarding the new police unit. The system must remain in a zero-trust state. This critical patch needs to be deployed across all production nodes before the end of the day."

Doing: I'm writing a Terraform script to automate the scaling of our backend microservices. I'm reviewing a pull request for a new chaincode deployment on the Hyperledger network. I'm running a security audit and penetration test on our new API endpoints. At 2 AM, I'm acknowledging a PagerDuty alert and SSH-ing into a server to diagnose a memory leak.

Hearing: I hear the constant ping-pong of Slack notifications for system alerts. I hear from the development team that they need new environment variables for the AI service. I hear from a project manager about the slow dashboard performance reported by a police unit. I hear the compliance team asking for audit logs to prove our data protection measures are working.

Feeling: I feel an immense sense of responsibility, knowing a system failure could have real-world consequences. I'm often stressed by the 24/7 on-call schedule and anxious about the unreliability of external dependencies we can't control. I feel frustrated when I have to debug a complex issue across microservices, but

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also a deep sense of satisfaction when I resolve it and see the system running smoothly and securely.

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